

App-03-Configure Application Stack

```
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.8) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
shourab@app01:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-08-05 14:00:26 UTC; 20s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3305 (apache2)
    Tasks: 55 (limit: 4545)
   Memory: 5.0M (peak: 5.3M)
      CPU: 29ms
   CGroup: /system.slice/apache2.service
           └─3305 /usr/sbin/apache2 -k start
             └─3307 /usr/sbin/apache2 -k start
               └─3308 /usr/sbin/apache2 -k start

Aug 05 14:00:26 app01 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 05 14:00:26 app01 apachectl[3304]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name
Aug 05 14:00:26 app01 systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-16/16 (END)
```

```

Setting up libhtml-template-perl (2.97-2) ...
Setting up mariadb-server (1:10.11.14-0ubuntu0.24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
Setting up mariadb-plugin-provider-bzip2 (1:10.11.14-0ubuntu0.24.04.1) ...
Setting up mariadb-plugin-provider-lzma (1:10.11.14-0ubuntu0.24.04.1) ...
Setting up mariadb-plugin-provider-lzo (1:10.11.14-0ubuntu0.24.04.1) ...
Setting up mariadb-plugin-provider-lz4 (1:10.11.14-0ubuntu0.24.04.1) ...
Setting up libcgi-fast-perl (1:2.17-1) ...
Setting up mariadb-plugin-provider-snappy (1:10.11.14-0ubuntu0.24.04.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.8) ...
Processing triggers for mariadb-server (1:10.11.14-0ubuntu0.24.04.1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
shourab@app01:~$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.11.14 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-08-05 14:09:00 UTC; 1min 20s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Main PID: 4415 (mariabdd)
    Status: "Taking your SQL requests now..."
      Tasks: 10 (limit: 29998)
    Memory: 80.4M (peak: 83.4M)
       CPU: 291ms
    CGroup: /system.slice/mariadb.service
            └─4415 /usr/sbin/mariabdd

Aug 05 14:09:00 app01 mariabdd[4415]: 2026-08-05 14:09:00 0 [Note] InnoDB: log sequence number 45502; transaction id 14
Aug 05 14:09:00 app01 mariabdd[4415]: 2026-08-05 14:09:00 0 [Note] Plugin 'FEEDBACK' is disabled.
Aug 05 14:09:00 app01 mariabdd[4415]: 2026-08-05 14:09:00 0 [Note] InnoDB: Loading buffer pool(s) from /var/lib/mysql/ib_buff
Aug 05 14:09:00 app01 mariabdd[4415]: 2026-08-05 14:09:00 0 [Warning] You need to use --log-bin to make --expire-logs-days or
Aug 05 14:09:00 app01 mariabdd[4415]: 2026-08-05 14:09:00 0 [Note] Server socket created on IP: '127.0.0.1', port: '3306'.
Aug 05 14:09:00 app01 mariabdd[4415]: 2026-08-05 14:09:00 0 [Note] InnoDB: Buffer pool(s) load completed at 260805 14:09:00
Aug 05 14:09:00 app01 mariabdd[4415]: 2026-08-05 14:09:00 0 [Note] /usr/sbin/mariabdd: ready for connections.
Aug 05 14:09:00 app01 mariabdd[4415]: Version: '10.11.14-MariaDB-0ubuntu0.24.04.1' socket: '/run/mysqld/mysqld.sock' port:
Aug 05 14:09:00 app01 systemd[1]: Started mariadb.service - MariaDB 10.11.14 database server.
Aug 05 14:09:00 app01 /etc/mysql/debian-start[4434]: Upgrading MariaDB tables if necessary.
shourab@app01:~$

```

● ● ●  shourab — shourab@app01: ~ — ssh shourab@172.16.229.150 — 99x33

```

● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-08-05 14:30:33 UTC; 1min 16s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 13656 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 13661 (apache2)
      Tasks: 6 (limit: 4545)
    Memory: 15.3M (peak: 15.6M)
       CPU: 54ms
    CGroup: /system.slice/apache2.service
            └─13661 /usr/sbin/apache2 -k start
              └─13665 /usr/sbin/apache2 -k start
                └─13666 /usr/sbin/apache2 -k start
                  └─13667 /usr/sbin/apache2 -k start
                    └─13668 /usr/sbin/apache2 -k start
                      └─13669 /usr/sbin/apache2 -k start

```

```

Aug 05 14:30:33 app01 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 05 14:30:33 app01 apachectl[13660]: AH00558: apache2: Could not reliably determine the server's
Aug 05 14:30:33 app01 systemd[1]: Started apache2.service - The Apache HTTP Server.

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```

lines 1-20/20 (END)

App-04-Configure MariaDB for Nextcloud

haven't set the root password yet, you should just press enter here.

```
[Enter current password for root (enter for none):  
OK, successfully used password, moving on...
```

Setting the root password or using the unix_socket ensures that nobody can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

```
[Switch to unix_socket authentication [Y/n] Y  
Enabled successfully!  
Reloading privilege tables..  
... Success!
```

You already have your root account protected, so you can safely answer 'n'.

```
[Change the root password? [Y/n] n  
... skipping.
```

By default, a MariaDB installation has an anonymous user, allowing anyone to log into MariaDB without having to have a user account created for them. This is intended only for testing, and to make the installation go a bit smoother. You should remove them before moving into a production environment.

```
[Remove anonymous users? [Y/n] Y  
... Success!
```

Normally, root should only be allowed to connect from 'localhost'. This ensures that someone cannot guess at the root password from the network.

```
[Disallow root login remotely? [Y/n] Y  
... Success!
```

By default, MariaDB comes with a database named 'test' that anyone can access. This is also intended only for testing, and should be removed before moving into a production environment.

```
[Remove test database and access to it? [Y/n] Y  
- Dropping test database...  
... Success!  
- Removing privileges on test database...  
... Success!
```

Reloading the privilege tables will ensure that all changes made so far will take effect immediately.

```
[Reload privilege tables now? [Y/n] Y  
... Success!
```

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB installation should now be secure.

Thanks for using MariaDB!

shourab@app01:~\$ █

```

shourab@app01:~$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 39
Server version: 10.11.14-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE nextcloud
-> CHARACTER SET utf8mb4
-> COLLATE utf8mb4_general_ci;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> Username:
-> nextclouduser
-> CREATE USER 'nextclouduser'@'localhost'
-> IDENTIFIED BY 'NcApp0102026Secure!';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'Username:
nextclouduser
CREATE USER 'nextclouduser'@'localhost'
IDENTIFIED BY...' at line 1
MariaDB [(none)]> CREATE USER 'nextclouduser'@'localhost'
-> IDENTIFIED BY 'NcApp0102026Secure!';
Query OK, 0 rows affected (0.009 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES
-> ON nextcloud.*
-> TO 'nextclouduser'@'localhost';
Query OK, 0 rows affected (0.008 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.003 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| nextcloud |
| performance_schema |
| sys |
+-----+
5 rows in set (0.008 sec)

MariaDB [(none)]> EXIT;
Bye
shourab@app01:~$

```

```

shourab@app01:~$ mysql -u nextclouduser -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 40
Server version: 10.11.14-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```

```

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| nextcloud |
+-----+
2 rows in set (0.006 sec)

```

```

MariaDB [(none)]> USE nextcloud;
Database changed
MariaDB [nextcloud]> EXIT;
Bye
shourab@app01:~$

```